$X^{s,b}$-SPACES

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1. Motivation

When studying solutions to scalar constant-coefficient dispersive linear equations (i.e. $\partial_t u = Lu$ for $L = ih(\nabla/i)$, h a real-valued polynomial), we saw the spacetime Fourier transform of free solutions, $\tilde{u}$, was supported entirely on the surface defined by dispersion relation $\tau = h(\xi)$ (see, e.g., section 2.1 of [4]). This algebraic property reflects the physical fact that, in a purely linear system, oscillations temporal and spatial domains are rigidly linked. If one considers adding a nonlinear perturbation to the equation, i.e.

$$\partial_t u = Lu + N(u), \tag{1}$$

this rigid support is lost. Intuitively, this nonlinearity can be thought of as something that spreads the spacetime Fourier transform of solutions away from the hypersurface $\tau = h(\xi)$. If the nonlinearity is strong or the initial datum is rough then one might expect the solution to lose its dispersive character entirely.

To study how this nonlinearity impacts the dispersive character of a solution, one might try to start their analysis using spaces we have already seen such as $C_t^0 H_x^s(\mathbb{R} \times \mathbb{R}^d)$ or $H_{t,x}^s(\mathbb{R} \times \mathbb{R}^d)$. Though helpful starting points, they do not fully reflect the structure of the underlying linear evolution. For example, $C_t^0 H_x^s$ views the solution as a family of snapshots in space and so does not directly encode the spacetime coupling imposed by dispersion, while $H_{t,x}^s$ measures isotropic regularity in (t, x) and thus ignores the specific geometry of the dispersion relation (i.e. it does not distinguish frequencies near the hypersurface $\tau = h(\xi)$). Hence, to better analyze nonlinear dispersive equations, we will first need a space of functions that is adapted to the specific dispersion relation of the equation.

Throughout this paper our domains are always $\mathbb{R}$ in the time variable and $\mathbb{R}^d$ in the spatial variable. Therefore, we often abbreviate $C_t^0 H_x^s \equiv C_t^0 H_x^s(\mathbb{R} \times \mathbb{R}^d)$ and analogous notation for different spaces for brevity.

The presentation of the material below generally follows the presentation from [4]. Following their notation, for a function f, we let $\hat{f}$ denote a Fourier transform in one variable (i.e. either spatial or temporal, the choice should be clear by the context) and $\tilde{f}$ denote a spacetime Fourier transform.

2. Definition and Basic Properties of $X^{s,b}$

Because of what was discussed above, we aim to describe some class of functions that still have the robust spatial regularity controls of standard Sobolev norms while introducing some weight that penalizes the "distance" in frequency space between the spacetime Fourier support and the dispersion relation hypersurface. This is achieved by weighting the spacetime Fourier transform with the modulation $\tau - h(x)$, thereby penalizing frequencies that lie far from the hypersurface $\tau = h(\xi)$. We encapsulate this in the following definition.

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Definition 1. *Let $h : \mathbb{R}^d \to \mathbb{R}$ be a continuous function and let $s, b \in \mathbb{R}$. The space $X^{s,b}_{\tau=h(\xi)}(\mathbb{R} \times \mathbb{R}^d) = X^{s,b}_{\tau=h(\xi)} = X^{s,b}$ is defined to be the closure of the Schwartz functions $\mathcal{S}_{t,x}(\mathbb{R} \times \mathbb{R}^d)$ under the norm:*

$$\|u\|_{X^{s,b}_{\tau=h(\xi)}(\mathbb{R}\times\mathbb{R}^d)} := \|\langle\xi\rangle^s \langle\tau - h(\xi)\rangle^b \tilde{u}(\tau,\xi)\|_{L^2_\tau L^2_\xi(\mathbb{R}\times\mathbb{R}^d)}. \tag{2}$$

These are referred to $X^{s,b}$-spaces (also known as Fourier restriction spaces, Bourgain spaces, or dispersive Sobolev spaces).

To make these spaces a bit less abstract, we relate them to familiar spaces in a few special cases. When $b = 0$, we see the following by Plancherel's theorem in time:

$$\|u\|_{X^{s,0}_{\tau=h(\xi)}(\mathbb{R}\times\mathbb{R}^d)} = \|\langle\xi\rangle^s \tilde{u}(\tau,\xi)\|_{L^2_\tau L^2_\xi(\mathbb{R}\times\mathbb{R}^d)} = \|u\|_{L^2_t H^s_x(\mathbb{R}\times\mathbb{R}^d)}, \tag{3}$$

i.e. $X^{s,0}_{\tau=h(\xi)}(\mathbb{R} \times \mathbb{R}^d) = L^2_t H^s_x(\mathbb{R} \times \mathbb{R}^d)$. When $h = 0$, we have

$$\|u\|_{X^{s,b}_{\tau=0}(\mathbb{R}\times\mathbb{R}^d)} = \|\langle\xi\rangle^s \langle\tau\rangle^b \tilde{u}(\tau,\xi)\|_{L^2_\tau L^2_\xi(\mathbb{R}\times\mathbb{R}^d)} = \|u\|_{H^b_t H^s_x(\mathbb{R}\times\mathbb{R}^d)}, \tag{4}$$

i.e. $X^{s,b}_{\tau=0}(\mathbb{R} \times \mathbb{R}^d) = H^b_t H^s_x(\mathbb{R} \times \mathbb{R}^d)$. We examine the relation between these two spaces for a general h in the lemma that follows.

Lemma 1. *Let $u \in \mathcal{S}(\mathbb{R} \times \mathbb{R}^d)$ be a complex field and $h : \mathbb{R}^d \to \mathbb{R}$ a polynomial. Let $U(t) := \exp(ith(\nabla/i))$ be the linear propagators for the equation $\partial_t u = Lu$ where $L = ih(\nabla/i)$ and let $v : \mathbb{R} \times \mathbb{R}^d \to \mathbb{C}$ be $v(t) := U(-t)u(t)$, i.e. v is constant in time if and only if u solves $\partial_t u = Lu$. Then for all $s, b \in \mathbb{R}$, we have*

$$\|u\|_{X^{s,b}_{\tau=h(\xi)}(\mathbb{R}\times\mathbb{R}^d)} = \|v\|_{H^b_t H^s_x(\mathbb{R}\times\mathbb{R}^d)}. \tag{5}$$

Remark 1. In words this says that $X^{s,b}_{\tau=h(\xi)}$ is a conjugate of $H^b_t H^s_x$. That is, we had some solution u and then we removed the linear oscillations to obtain v so that the dispersive modulation, $\tau - h(\xi)$, just becomes time frequency, τ. Thus, the $X^{s,b}$ norm just becomes the standard Sobolev norm after factoring out the linear phase $e^{ith(\xi)}$. This plays a fundamental role in the breakdown we observe in Section 5.

Proof. Writing $U(t) = e^{tL}$, we recall $\widehat{e^{tL} f}(\xi) = e^{ith(\xi)} \hat{f}(\xi)$. Hence

$$v(t) = U(-t)u(t) = e^{-tL}u(t) \implies \widetilde{v}(\tau,\xi) = \widetilde{u}(\tau + h(\xi), \xi), \tag{6}$$

since the complex exponential corresponds to a frequency shift when taking the temporal Fourier transform. Equivalently, we obtain $\widetilde{u}(\tau,\xi) = \widetilde{v}(\tau - h(\xi), \xi)$. Now taking the $X^{s,b}$ norm:

$$\|u\|^2_{X^{s,b}_{\tau=h(\xi)}} = \int_{\mathbb{R}^d} \int_{\mathbb{R}} \langle\xi\rangle^{2s} \langle\tau - h(\xi)\rangle^{2b} |\tilde{u}(\tau,\xi)|^2 d\tau d\xi = \int_{\mathbb{R}^d} \int_{\mathbb{R}} \langle\xi\rangle^{2s} \langle\sigma\rangle^{2b} |\tilde{v}(\sigma,\xi)|^2 d\sigma d\xi = \|v\|^2_{H^b_t H^s_x}, \tag{7}$$

where we did the change of variables $\sigma = \tau - h(\xi)$. Thus we have shown $\|u\|_{X^{s,b}_{\tau=h(\xi)}(\mathbb{R}\times\mathbb{R}^d)} = \|v\|_{H^b_t H^s_x(\mathbb{R}\times\mathbb{R}^d)}$. □

Before turning to nonlinear dispersive equations, we will first verify that these spaces are compatible with the linear evolution. In particular, we guarantee that time-localized free solutions (i.e. solutions to the linear dispersive equation $\partial_t u = Lu$ multiplied by a Schwartz time cutoff) lie in $X^{s,b}$ with norms controlled by the initial data. We formulate this in the following lemma.

Lemma 2. *Let $f \in H^s_x(\mathbb{R}^d)$ for some $s \in \mathbb{R}$ and $L = ih(\nabla/i)$ where $h : \mathbb{R}^d \to \mathbb{R}$ is a polynomial. Then for any Schwartz time cutoff, $\eta \in \mathcal{S}_t(\mathbb{R})$, we obtain*

$$\|\eta(t) e^{tL} f\|_{X^{s,b}_{\tau=h(\xi)}(\mathbb{R}\times\mathbb{R}^d)} \lesssim_{\eta,b} \|f\|_{H^s_x(\mathbb{R}^d)}. \tag{8}$$

Remark 2. Thus, time-localized solutions to the linear equation $\partial_t u = Lu$ with initial data in $H^s_x(\mathbb{R}^d)$ lie in $X^{s,b}_{\tau=h(\xi)}$, since $u(t) = e^{tL}f$.

Proof. The proof of this statement follows from computing the spacetime Fourier transform of $\eta(t)e^{tL}f$ at (τ, ξ), i.e.:

$$\widehat{e^{tL}f}(\xi) = e^{ith(\xi)}\hat{f}(\xi) \implies \widetilde{\eta e^{tL}f}(\tau,\xi) = \hat{f}(\xi)\int_{\mathbb{R}} \eta(t)e^{ith(\xi)}e^{-it\tau}dt \tag{9}$$

$$= \hat{f}(\xi)\int_{\mathbb{R}} \eta(t)e^{it(h(\xi)-\tau)}dt = \hat{f}(\xi)\hat{\eta}(\tau - h(\xi)). \tag{10}$$

Using this to compute the norm:

$$\|\eta(t)e^{tL}f\|^2_{X^{s,b}} = \int_{\mathbb{R}^d}\int_{\mathbb{R}} \langle\xi\rangle^{2s}\langle\tau - h(\xi)\rangle^{2b}|\hat{\eta}(\tau - h(\xi))|^2|\hat{f}(\xi)|^2 d\tau d\xi \tag{11}$$

$$= \left(\int_{\mathbb{R}} \langle\sigma\rangle^{2b}|\hat{\eta}(\sigma)|^2 d\sigma\right)\left(\int_{\mathbb{R}^d}\langle\xi\rangle^{2s}|\hat{f}(\xi)|^2 d\xi\right) \lesssim_{\eta,b} \int_{\mathbb{R}^d}\langle\xi\rangle^{2s}|\hat{f}(\xi)|^2 d\xi = \|f\|^2_{H^s_x(\mathbb{R}^d)}, \tag{12}$$

where the inequality follows from the fact that η Schwartz implies that $\hat{\eta}$ is Schwartz and so decays rapidly (i.e. decays more quickly than any polynomial grows). □

These observations show that $X^{s,b}$ is well suited to the linear evolution. The next step is to see how these general linear estimates transfer to the full space.

3. Linear Estimates in $X^{s,b}$

A key utility of $X^{s,b}$ spaces is that these estimates for free solutions extend to general functions in $X^{s,b}$. That is, the estimates of Lemma 2 only applied to free solutions of the linear equation and their time-localized cutoffs. To handle the nonlinear equation, however, we first need these estimates to hold for *arbitrary* elements of $X^{s,b}$, not just those of the form $\eta e^{tL}f$. The following lemma gives this extension.

Lemma 3. *Let $L = iP(\nabla/i)$ for some polynomial $P : \mathbb{R}^d \to \mathbb{R}$, let $s \in \mathbb{R}$ and let Y be a Banach space on $\mathbb{R}\times\mathbb{R}^d$ satisfying*

$$\|e^{it\tau_0}e^{tL}f\|_Y \lesssim \|f\|_{H^s_x(\mathbb{R}^d)} \quad \forall\ f \in H^s_x(\mathbb{R}^d), \tau_0 \in \mathbb{R}. \tag{13}$$

Then for $b > 1/2$, $u \in X^{s,b}$ we have the embedding

$$\|u\|_Y \lesssim_b \|u\|_{X^{s,b}}. \tag{14}$$

Remark 3. This lemma allows us to pass from estimates for free solutions to estimates for general elements of $X^{s,b}$. Indeed, Lemma 2 shows that functions of the form $\eta(t)e^{tL}f$ obey the desired bounds. This lemma asserts that, under suitable conditions, every $u \in X^{s,b}$ can be controlled by such free evolutions, and hence satisfies the same estimates. In this sense, this lemma upgrades Lemma 2 from time-localized free solutions to arbitrary elements of $X^{s,b}$.

Proof. By Fourier inversion, we can write

$$u(t,x) = \frac{1}{(2\pi)^{d+1}}\int_{\mathbb{R}}\int_{\mathbb{R}^d} \tilde{u}(\tau,\xi)e^{it\tau}e^{ix\cdot\xi}d\xi d\tau. \tag{15}$$

Thus, we write $\tau = h(\xi) + \tau_0$ and define

$$f_{\tau_0}(x) := \frac{1}{(2\pi)^d}\int_{\mathbb{R}^d}\tilde{u}(h(\xi)+\tau_0,\xi)e^{ix\cdot\xi}d\xi \implies \|f_{\tau_0}\|^2_{H^s_x(\mathbb{R}^d)} = \int_{\mathbb{R}^d}\langle\xi\rangle^{2s}|\tilde{u}(h(\xi)+\tau_0,\xi)|^2 d\xi, \tag{16}$$

Using $\widehat{e^{tL}f}(\xi) = e^{ith(\xi)}\hat{f}(\xi)$, we see

$$u(t) = \frac{1}{2\pi}\int_{\mathbb{R}} e^{it\tau_0}e^{tL}f_{\tau_0}d\tau_0. \tag{17}$$

Hence by a triangle inequality and Cauchy-Schwarz using $b > 1/2$ by assumption

$$\|u\|_Y \lesssim \int_{\mathbb{R}} \|f_{\tau_0}\|_{H^s_x(\mathbb{R}^d)}d\tau_0 \leq \left(\int_{\mathbb{R}} \langle\tau_0\rangle^{-2b}d\tau_0\right)^{1/2}\left(\int_{\mathbb{R}} \langle\tau_0\rangle^{2b}\|f_{\tau_0}\|^2_{H^s_x(\mathbb{R}^d)}d\tau_0\right)^{1/2} \tag{18}$$

$$\lesssim_b \left(\int_{\mathbb{R}} \langle\tau_0\rangle^{2b}\|f_{\tau_0}\|^2_{H^s_x(\mathbb{R}^d)}d\tau_0\right)^{1/2} \tag{19}$$

$$= \left(\int_{\mathbb{R}}\int_{\mathbb{R}^d} \langle\tau - h(\xi)\rangle^{2b}\langle\xi\rangle^{2s}|\tilde{u}(\tau,\xi)|^2 d\xi d\tau\right)^{1/2} \tag{20}$$

$$= \left\|\langle\tau - h(\xi)\rangle^b\langle\xi\rangle^s\tilde{u}\right\|_{L^2_\tau L^2_\xi} = \|u\|_{X^{s,b}}, \tag{21}$$

where the final equality follows from recalling the definition of the $X^{s,b}$ norm. □

This lemma really highlights the power of these $X^{s,b}$-spaces, as demonstrated in the two corollaries below.

Corollary 1. *Let $b > 1/2$, $s \in \mathbb{R}$, and $h : \mathbb{R}^d \to \mathbb{R}$ be continuous. Then for any $u \in X^{s,b}_{\tau=h(\xi)}(\mathbb{R}\times\mathbb{R}^d)$ we have*

$$\|u\|_{C^0_t H^s_x(\mathbb{R}\times\mathbb{R}^d)} \lesssim_b \|u\|_{X^{s,b}_{\tau=h(\xi)}(\mathbb{R}\times\mathbb{R}^d)}. \tag{22}$$

Proof. We are justified in applying Lemma 3 with $Y = C^0_t H^s_x(\mathbb{R}\times\mathbb{R}^d)$ and $L = ih(\nabla/i)$ since $\widehat{e^{tL}f}(\xi) = e^{ith(\xi)}\hat{f}(\xi)$ for any $f \in H^s_x$. Hence, using the fact that for any $\tau_0 \in \mathbb{R}$, $|e^{ith(\xi)}| = |e^{it\tau_0}| = 1$, we see

$$\|e^{it\tau_0}e^{tL}f\|_{H^s_x} = \|f\|_{H^s_x} \implies \|e^{it\tau_0}e^{tL}f\|_{C^0_t H^s_x} = \sup_t \|e^{it\tau_0}e^{tL}f\|_{H^s_x} = \|f\|_{H^s_x}, \tag{23}$$

which is the bound needed to apply the lemma. □

Corollary 2 (Strichartz estimate)**.** *Let $d \geq 1$ and call a pair of exponents (q,r) admissible if $2 \leq q, r \leq \infty$ satisfy $\frac{2}{q} + \frac{d}{r} = \frac{d}{2}$ and $(q,r,d) \neq (2,\infty,2)$. Let $h(\xi) = -|\xi|^2/2$ and $b > 1/2$. Then for any admissible exponents (q,r) we have*

$$\|u\|_{L^q_t L^r_x(\mathbb{R}\times\mathbb{R}^d)} \lesssim_{d,q,r,b} \|u\|_{X^{0,b}_{\tau=h(\xi)}(\mathbb{R}\times\mathbb{R}^d)}. \tag{24}$$

Remark 4. This is a very powerful statement since the homogeneous Strichartz estimate (eq. (26)) only worked for linear evolution.

Proof. We first note the following

$$h(\xi) = -|\xi|^2/2 \implies h(\nabla/i) = -\frac{1}{2}|\nabla/i|^2 = -\frac{1}{2}(-i\nabla)\cdot(-i\nabla) = -\frac{1}{2}(-i)^2\nabla^2 = \Delta/2. \tag{25}$$

Thus, we want to apply Lemma 3 with $Y = L^q_t L^r_x(\mathbb{R}\times\mathbb{R}^d)$, $s = 0$, and $L = \frac{i}{2}\Delta$, so that $e^{tL} = e^{it\Delta/2}$. To justify applying this we first note that for any $f \in H^0_x(\mathbb{R}^d)$,

$$\|e^{it\Delta/2}f\|_{L^q_t L^r_x} \lesssim_{d,q,r} \|f\|_{L^2_x} \tag{26}$$

is precisely the homogeneous Strichartz estimate (see, e.g., equation 2.24 of [4]). So after noting that $\|f\|_{L^2_x} = \|f\|_{H^0_x}$ and that for any $\tau_0 \in \mathbb{R}$ the complex exponential has norm 1, we obtain

$$\|e^{it\tau_0}e^{it\Delta/2}f\|_{L^q_t L^r_x} = \|e^{it\Delta/2}f\|_{L^q_t L^r_x} \lesssim_{d,q,r} \|f\|_{L^2_x} = \|f\|_{H^0_x}. \tag{27}$$

Lemma 3 then yields

$$\|u\|_{L_t^q L_x^r} \lesssim_{d,q,r,b} \|u\|_{X^{0,b}_{\tau=h(\xi)}}. \tag{28}$$

□

4. Nonlinear Local-In-Time Estimates for $X^{s,b}$

We now turn to the central application of the theory developed above: using $X^{s,b}$ spaces to analyze nonlinear dispersive equations of the form eq. (1). Our general approach is to reduce the problem to an inhomogeneous linear equation $\partial_t u = Lu + F$ by treating the nonlinear term $N(u)$ as a forcing term F. We run into an immediate problem, however, since $X^{s,b}$ is defined globally in time but controlling a nonlinear term typically requires working on a sufficiently short time interval so that the solution does not deviate too far from its initial value (see the proof of Proposition 1 for why this is a problem). One must therefore restrict the analysis to a short time interval, which requires localizing functions in $X^{s,b}$ by multiplying by a smooth time cutoff. Luckily, these spaces are easy to localize, as we demonstrate in the following lemma.

Lemma 4. *Let $\eta \in \mathcal{S}_t(\mathbb{R})$ be a Schwartz time cutoff. Then for any $s, b \in \mathbb{R}$, $h : \mathbb{R}^d \to \mathbb{R}$, and $u \in \mathcal{S}_{t,x}(\mathbb{R} \times \mathbb{R}^d)$ a field, we have*

$$\|\eta(t)u\|_{X^{s,b}_{\tau=h(\xi)}(\mathbb{R}\times\mathbb{R}^d)} \lesssim_{\eta,b} \|u\|_{X^{s,b}_{\tau=h(\xi)}(\mathbb{R}\times\mathbb{R}^d)}. \tag{29}$$

Proof. Let us begin by investigating how the $X^{s,b}$ norm is impacted by a temporal frequency modulation, i.e. $u(t,x) \mapsto \widetilde{e^{it\tau_0}u}(t,x)$ for $\tau_0 \in \mathbb{R}$. By an elementary property of the Fourier transform, we of course have $\widetilde{e^{it\tau_0}u}(\tau,\xi) = \tilde{u}(\tau - \tau_0, \xi)$ and so by the elementary estimate $\langle \tau + \tau_0 - h\rangle^b \lesssim_b \langle \tau_0\rangle^{|b|}\langle \tau - h\rangle^b$, we obtain

$$\|e^{it\tau_0}u\|_{X^{s,b}} = \|\langle\xi\rangle^s \langle \tau - h(\xi)\rangle^b \tilde{u}(\tau - \tau_0, \xi)\|_{L^2_\tau L^2_\xi} \tag{30}$$

$$= \|\langle\xi\rangle^s \langle \tau + \tau_0 - h(\xi)\rangle^b \tilde{u}(\tau, \xi)\|_{L^2_\tau L^2_\xi} \lesssim_b \langle\tau_0\rangle^{|b|}\|u\|_{X^{s,b}}, \tag{31}$$

where the second equality followed from doing the change of variables $\tau \mapsto \tau + \tau_0$ in the integral when computing $\|\cdot\|_{L^2_\tau L^2_\xi}$. Hence, writing $\eta(t) = \int_{\mathbb{R}} \hat{\eta}(\tau_0) e^{it\tau_0} d\tau_0$ by a Fourier expansion, we see by an elementary triangle inequality and the above bound

$$\|\eta(t)u\|_{X^{s,b}} \leq \int_{\mathbb{R}} |\hat{\eta}(\tau_0)| \|e^{it\tau_0}u\|_{X^{s,b}} d\tau_0 \lesssim_b \left(\int_{\mathbb{R}} \langle\tau_0\rangle^{|b|} |\hat{\eta}(\tau_0)| d\tau_0\right) \|u\|_{X^{s,b}}. \tag{32}$$

But since η is Schwartz, so is its Fourier transform, implying that $\langle\tau_0\rangle^{|b|}|\hat{\eta}(\tau_0)|$ is integrable. Hence, we obtain the bound $\|\eta(t)u\|_{X^{s,b}} \lesssim_{\eta,b} \|u\|_{X^{s,b}}$. □

With time localization taken care of, we can now formulate the key linear estimate for the inhomogeneous problem.

Proposition 1. *Let $h : \mathbb{R}^d \to \mathbb{R}$ be a polynomial, $L := ih(\nabla/i)$, and $u \in C^\infty_{t,loc}\mathcal{S}_x(\mathbb{R} \times \mathbb{R}^d)$ be a solution to $\partial_t u = Lu + F$. Then for any $s \in \mathbb{R}$ & $b > 1/2$ and any smooth time cutoff $\eta(t)$ with compact support, we have*

$$\|\eta(t)u\|_{X^{s,b}_{\tau=h(\xi)}(\mathbb{R}\times\mathbb{R}^d)} \lesssim_{\eta,b} \|u(0)\|_{H^s_x(\mathbb{R}^d)} + \|F\|_{X^{s,b-1}_{\tau=h(\xi)}(\mathbb{R}\times\mathbb{R}^d)}. \tag{33}$$

Remark 5. This is what makes $X^{s,b}$ spaces so effective for nonlinear problems. It shows that the solution u is controlled in $X^{s,b}$ by the initial datum in H^s_x and the forcing term F in $X^{s,b-1}$. Taking $F = N(u)$, this proposition reduces analyzing eq. (1) to proving an estimate for the nonlinearity $N(u)$ in an $X^{s,b-1}$ norm.

Proof. Choose R large enough such that $\operatorname{supp}\eta \subseteq [-R,R]$. By smoothly truncating F to $[-2R,2R]$ using a cutoff that is 1 on $[-R,R]$ we may assume without loss of generality that F is supported on $[-2R,2R]$ and Schwartz in spacetime by Lemma 4.

We note that $\langle\nabla\rangle^s$ commutes with scalars dependent on time (e.g. η) and it commutes with $L = ih(\nabla/i)$ since they are both Fourier multipliers in x with constant coefficients. Therefore, we can set $s = 0$ without loss of generality. That is if we prove the bound for $s = 0$ we can apply it for $v = \langle\nabla\rangle^s u$ to obtain it for an arbitrary s. After this reduction, the desired bound becomes

$$\|\eta(t)u\|_{X^{0,b}_{\tau=h(\xi)}} \lesssim_{\eta,b} \|u(0)\|_{H^0_x} + \|F\|_{X^{0,b-1}_{\tau=h(\xi)}} = \|u(0)\|_{L^2_x} + \|F\|_{X^{0,b-1}_{\tau=h(\xi)}}. \tag{34}$$

First, let's consider the case where $u(-2R) = 0$. Then, by Duhamel's formula (see, e.g., equation 2.13 of [4]) with $t_0 = -2R$, we obtain

$$u(t) = e^{(t+2R)L}u(-2R) + \int_{-2R}^{t} e^{(t-s)L}F(s)ds = \int_{-\infty}^{t} e^{(t-s)L}F(s)ds, \tag{35}$$

since $F(s) = 0$ for $s < -2R$. Then, introducing a smooth cutoff $\tilde\eta$ that is 1 on $[-3R,3R]$ so that $\tilde\eta(t-s) = 1$ for any $(t,s) \in \operatorname{supp}\eta(t)F(s)$, we see

$$\eta(t)u(t) = \eta(t)\int_{\mathbb{R}} \tilde\eta(t-s)\chi_{[0,\infty)}(t-s)e^{(t-s)L}F(s)ds, \tag{36}$$

where $\chi_{[0,\infty)}$ is the characteristic function. Hence, by Lemma 4

$$\|\eta(t)u(t)\|_{X^{0,b}} = \left\|\eta(t)\int_{\mathbb{R}} \tilde\eta(t-s)\chi_{[0,\infty)}(t-s)e^{(t-s)L}F(s)ds\right\|_{X^{0,b}} \tag{37}$$

$$\lesssim_{\eta,b} \left\|\int_{\mathbb{R}} \tilde\eta(t-s)\chi_{[0,\infty)}(t-s)e^{(t-s)L}F(s)ds\right\|_{X^{0,b}}. \tag{38}$$

Thus, because $\|u(0)\|_{L^2_x} \geq 0$, it suffices to prove the bound

$$\left\|\int_{\mathbb{R}} \tilde\eta(t-s)\chi_{[0,\infty)}(t-s)e^{(t-s)L}F(s)ds\right\|_{X^{0,b}} \lesssim_{\eta,b} \|F\|_{X^{0,b-1}}. \tag{39}$$

We now want to take the spacetime Fourier transform of this integral. Recalling eq. (9)

$$\mathcal{F}_x\left(\int_{\mathbb{R}} \tilde\eta(t-s)\chi_{[0,\infty)}(t-s)e^{(t-s)L}F(s)ds\right)(t,\xi) = \int_{\mathbb{R}} \tilde\eta(t-s)\chi_{[0,\infty)}(t-s)e^{i(t-s)h(\xi)}\widehat{F}(s,\xi)ds. \tag{40}$$

Recognizing this as a convolution of $\widehat{F}$ with the kernel $\tilde\eta(s)\chi_{[0,\infty)}(s)e^{ish(\xi)}$, recalling that convolutions become multiplication after taking the Fourier transform, we see that the spacetime Fourier transform becomes

$$\mathcal{F}_{x,t}\left(\int_{\mathbb{R}} \tilde\eta(t-s)\chi_{[0,\infty)}(t-s)e^{(t-s)L}F(s)ds\right)(\tau,\xi) = m(\tau - h(\xi))\widetilde{F}(\tau,\xi), \tag{41}$$

where we set $m(\omega) := \int_{\mathbb{R}} \tilde\eta(t)\chi_{[0,\infty)}(t)e^{-it\omega}dt$. Performing an elementary integration by parts, using that $\tilde\eta$ is compactly supported, we obtain

$$|m(\omega)| \lesssim_\eta \min(1, |\omega|^{-1}) \lesssim \langle\omega\rangle^{-1}, \tag{42}$$

where the first bound follows from treating the cases $|\omega| \leq 1$ and $|\omega| > 1$ separately. Hence the claim follows since

(43)
$$\left\| \int_{\mathbb{R}} \tilde{\eta}(t-s)\chi_{[0,\infty)}(t-s)e^{(t-s)L}F(s)ds \right\|_{X^{0,b}}^2 \lesssim_\eta \int_{\mathbb{R}} \int_{\mathbb{R}^d} \langle \tau - h(\xi)\rangle^{2b} \langle \tau - h(\xi)\rangle^{-2} |\widetilde{F}(\tau,\xi)|^2 d\tau d\xi$$

(44)
$$= \int_{\mathbb{R}} \int_{\mathbb{R}^d} \langle \tau - h(\xi)\rangle^{2(b-1)} |\widetilde{F}(\tau,\xi)|^2 d\tau d\xi = \|F\|_{X^{0,b-1}}^2.$$

We now proceed to the general case. To begin, let's rewrite u as

$$u(t) = (u(t) - e^{(t+2R)L}u(-2R)) + e^{(t+2R)L}u(-2R). \tag{45}$$

Because the first term equals 0 when $t = -2R$, we can perform the above analysis to get the desired bound. Hence it suffices to show the bound on the second term. By Lemma 2, we have

$$\|\eta(t)e^{(t+2R)L}u(-2R)\|_{X^{0,b}} \lesssim_{\eta,b} \|u(-2R)\|_{L^2_x}. \tag{46}$$

By Duhamel's formula at time $-2R < 0$, we obtain

$$e^{2RL}u(-2R) = u(0) - \int_{\mathbb{R}} \tilde{\eta}(s)\chi_{(-\infty,0]}(s)e^{-sL}F(s)ds, \tag{47}$$

with $\tilde{\eta}$ as above. Thus, because we have a $\|u(0)\|_{L^2_x}$ term in the final bound (i.e. since the L^2 and H^0 norm of u are the same), a triangle inequality shows that the following bound suffices

$$\left\| \int_{\mathbb{R}} \tilde{\eta}(s)\chi_{(-\infty,0]}(s)e^{-sL}F(s)ds \right\|_{L^2_x} \lesssim_{\eta,b} \|F\|_{X^{0,b-1}}. \tag{48}$$

By applying Plancherel's theorem in space we see

$$\left\| \int_{\mathbb{R}} \tilde{\eta}(s)\chi_{(-\infty,0]}(s)e^{-sL}F(s)ds \right\|_{L^2_x} = \left\| \int_{\mathbb{R}} \left(\int_{\mathbb{R}} \tilde{\eta}(s)\chi_{(-\infty,0]}(s)e^{is(\tau-h(\xi))}ds \right) \widehat{F}(\tau,\xi)d\tau \right\|_{L^2_\xi} \tag{49}$$

As argued in eq. (42), an integration by parts and triangle inequality gives

$$\left| \int_{\mathbb{R}} \left(\int_{\mathbb{R}} \tilde{\eta}(s)\chi_{(-\infty,0]}(s)e^{is(\tau-h(\xi))}ds \right) \widetilde{F}(\tau,\xi)d\tau \right| \lesssim_\eta \int_{\mathbb{R}} \langle \tau - h(\xi)\rangle^{-1} |\widetilde{F}(\tau,\xi)|d\tau. \tag{50}$$

So by a Cauchy-Schwarz in τ,

(51)
$$\int_{\mathbb{R}} \langle \tau - h(\xi)\rangle^{-1} |\widetilde{F}(\tau,\xi)|d\tau = \int_{\mathbb{R}} \langle \tau - h(\xi)\rangle^{-b} \left(\langle \tau - h(\xi)\rangle^{b-1} |\widetilde{F}(\tau,\xi)| \right) d\tau$$

(52)
$$\leq \left(\int_{\mathbb{R}} \langle \tau - h(\xi)\rangle^{-2b} d\tau \right)^{1/2} \left(\int_{\mathbb{R}} \langle \tau - h(\xi)\rangle^{2(b-1)} |\widetilde{F}(\tau,\xi)|^2 d\tau \right)^{1/2}.$$

Elementary analysis shows the the first factor is finite iff $2b > 1 \iff b > 1/2$ (i.e. compare the integral and sum). Further since we integrate over all τ, the value of the first factor is independent of ξ. Hence, squaring integrating over ξ, we obtain

$$\left\| \int_{\mathbb{R}} \tilde{\eta}(s)\chi_{(-\infty,0]}(s)e^{-sL}F(s)ds \right\|_{L^2_x}^2 \lesssim_{\eta,b} \int_{\mathbb{R}^d} \int_{\mathbb{R}} \langle \tau - h(\xi)\rangle^{2(b-1)} |\widetilde{F}(\tau,\xi)|^2 d\tau d\xi = \|F\|_{X^{0,b-1}}^2. \tag{53}$$

Thus, square rooting gives us eq. (48), proving the statement in the general case. □

5. A Failure Case and Further Directions

Proposition 2. *The conclusions of Lemma 3 and Corollary 1 cannot be extended to the endpoint* $b = 1/2$.

Proof. We will first construct a counterexample for Corollary 1 when $b = 1/2$. Take any nonzero $\phi \in C_c^\infty(\mathbb{R}^d)$ (i.e. smooth with compact support), let $\chi_{[a,b]}(x)$ be the characteristic function of the interval $[a, b]$, and for $N \geq 2$, define u_N in terms of its spacetime Fourier transform

$$\widetilde{u_N}(\tau, \xi) := \langle \xi \rangle^{-s} \phi(\xi) \psi_N(\tau - h(\xi)) \quad \text{where } \psi_N(\omega) = \frac{1}{\sqrt{\log N}} \frac{\chi_{[1,N]}(\omega)}{\omega}. \tag{54}$$

Hence

$$\|u_N\|^2_{X^{s,1/2}} = \int_{\mathbb{R}^d} |\phi(\xi)|^2 \left(\int_1^N \frac{\langle \tau \rangle}{\tau^2 \log N} d\tau \right) d\xi \lesssim \|\phi\|^2_{L^2_\xi} < \infty \implies u_N \in X^{s,1/2}_{\tau = h(\xi)}(\mathbb{R} \times \mathbb{R}^d). \tag{55}$$

At $t = 0$, however, we see

$$\widehat{u_N(0, \cdot)}(\xi) = \int_{\mathbb{R}} \hat{u}_N(\tau, \xi) d\tau = \langle \xi \rangle^{-s} \phi(\xi) \frac{1}{\sqrt{\log N}} \int_1^N \frac{d\tau}{\tau} = \langle \xi \rangle^{-s} \phi(\xi) \sqrt{\log N}. \tag{56}$$

So we have shown

$$\|u\|^2_{C^0_t H^s_x} \geq \|u_N(0, \cdot)\|^2_{H^s_x} = \int_{\mathbb{R}^d} \langle \xi \rangle^{2s} |\widehat{u_N(0, \cdot)}(\xi)|^2 d\xi = \|\phi\|^2_{L^2_\xi} \log N \to \infty \text{ as } N \to \infty. \tag{57}$$

That is, there does not exist a constant M such that

$$\|u\|_{C^0_t H^s_x(\mathbb{R} \times \mathbb{R}^d)} \leq M \|u\|_{X^{s,1/2}_{\tau = h(\xi)}(\mathbb{R} \times \mathbb{R}^d)} \quad \forall\ u \in X^{s,1/2}_{\tau = h(\xi)}(\mathbb{R} \times \mathbb{R}^d), \tag{58}$$

since, e.g., $u_N \in X^{s,1/2}_{\tau = h(\xi)}(\mathbb{R} \times \mathbb{R}^d)$ and $\|u_N\|_{C^0_t H^s_x(\mathbb{R} \times \mathbb{R}^d)}$ grows without bound as $N \to \infty$ while $\|u_N\|_{X^{s,1/2}_{\tau = h(\xi)}(\mathbb{R} \times \mathbb{R}^d)}$ remains uniformly bounded.

This example is clearly also a counterexample to Lemma 3. □

This demonstrates that at the point $b = 1/2$, the behavior of these spaces changes drastically. To understand this breakdown, we will state the Sobolev embedding theorem, the proof of which is a standard result from functional analysis (see, e.g., equation A.13 and the surrounding text in [4] in the case $d = 1$).

Theorem 1 (Sobolev embedding)**.** *The bound*

$$\|f\|_{L^\infty(\mathbb{R})} \lesssim_b \|f\|_{H^b(\mathbb{R})} \tag{59}$$

holds for all $f \in H^b(\mathbb{R})$ *iff* $b > 1/2$.

Recalling Lemma 1, this implies the breakdown since, following the notation of that lemma, we have

$$\|v\|_{C^0_t H^s_x} = \|v\|_{L^\infty_t H^s_x(\mathbb{R} \times \mathbb{R}^d)} \lesssim_b \|v\|_{H^b_t H^s_x(\mathbb{R} \times \mathbb{R}^d)} \quad \text{iff } b > 1/2. \tag{60}$$

and therefore

$$\|u\|_{C^0_t H^s_x} = \sup_{t \in \mathbb{R}} \|u(t)\|_{H^s_x} = \sup_{t \in \mathbb{R}} \|v(t)\|_{H^s_x} = \|v\|_{L^\infty_t H^s_x} \lesssim_b \|v\|_{H^b_t H^s_x} = \|u\|_{X^{s,b}} \quad \text{iff } b > 1/2, \tag{61}$$

showing that the condition $b > 1/2$ in Corollary 1 is sharp.

For $b \leq 1/2$, one often replaces $X^{s,b}$ by atomic time-frequency spaces such as the U^p and V^p frameworks, which are better suited for low-regularity nonlinear estimates. These are better suited because the atomic spaces encode time evolution through piecewise constant or bounded-variation structure along the linear evolution. These frameworks can retain good control of nonlinear evolutions at or even below the $b = 1/2$ threshold, where $X^{s,b}$ no longer embeds into $C^0_t H^s_x$. For further

reading on these approaches we refer the reader to [2] to learn more about the origin of these atomic spaces and [3] for an interesting (though rather technical) application of these spaces to analyze the nonlinear Schrödinger equation.

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